

Lecture 6B - Quasi-Experimental Designs

Regression Discontinuity Design

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Outline

- 1 Getting a little “jumpy”: the basics of RDD
- 2 RDD: A Formal Examination
 - The data-generating process of RDD
 - Local experiment perspective
 - Potential-outcomes perspective
- 3 Estimation Issues for RDD
 - Predict conditional expectations
 - Select appropriate samples
- 4 RDD Validation
- 5 Other variants of RDD
- 6 Slides for R Notebook

- ## ① Getting a little “jumpy”: the basics of RDD

- ## ② RDD: A Formal Examination

- ### ③ Estimation Issues for RDD

- #### 4 RDD Validation

- ## 5 Other variants of RDD

- ## 6 Slides for R Notebook

The experiment we would like to run—but cannot

Ideal experiment

Randomly assign cum laude honors among students with similar academic performance, then compare their later earnings.

Problem

Universities cannot randomly award academic honors.

Opportunity

The university already uses a rule:

$$D_i = \begin{cases} 1, & \text{if GPA}_i \geq 8.25 \\ 0, & \text{if GPA}_i < 8.25 \end{cases}$$

Can cum laude honors increase earnings?

Comparison 1

- Student A: GPA = 8.24, no cum laude
- Student B: GPA = 8.26, cum laude

Comparison 2

- Student C: GPA = 5.50, no cum laude
- Student D: GPA = 8.50, cum laude

Which comparison is better?

Does the comparison of close students give us the causal effect?

- Student A: GPA = 8.24, no cum laude
- Student B: GPA = 8.26, cum laude

- Student C: GPA = 5.50, no cum laude
- Student D: GPA = 8.50, cum laude

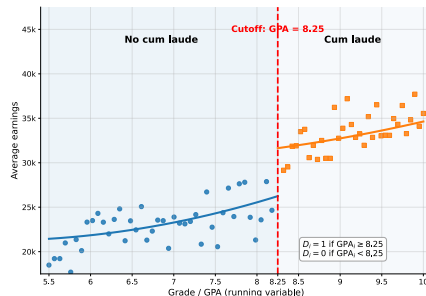
Which comparison is better?

Does the comparison of close students give us the causal effect?

A rule creates treatment assignment

- The **running variable** is GPA.
- The **cutoff** is 8.25.
- The **treatment** is cum laude honors.
- Treatment changes discontinuously at the cutoff.

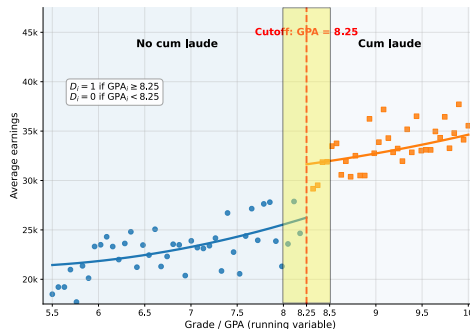
The rule creates a sharp change in treatment status.



RDD compares near misses and narrow winners

Experimental-design intuition

Near the cutoff, treatment assignment may be **as-good-as-random**.



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Business settings often contain rules, ranks, ratings, and thresholds.

Displayed stars affect demand

Effect operates through assignment at the cutoff

Visibility affects consumption

Effect operates through assignment at the cutoff

Benefits affect future purchases

Effect operates through assignment at the cutoff

Index assignment affects ownership

Effect operates through assignment at the cutoff

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A business example: platform star ratings

RDD in review platforms

- **Running variable:** average review score
- **Cutoff:** rounding threshold
- **Treatment:** displayed star rating
- **Outcome:** product sales

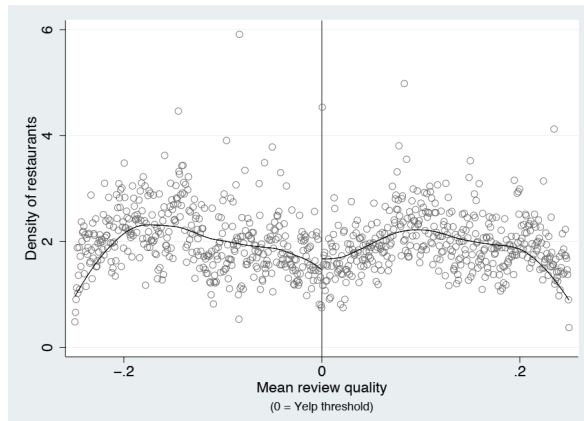
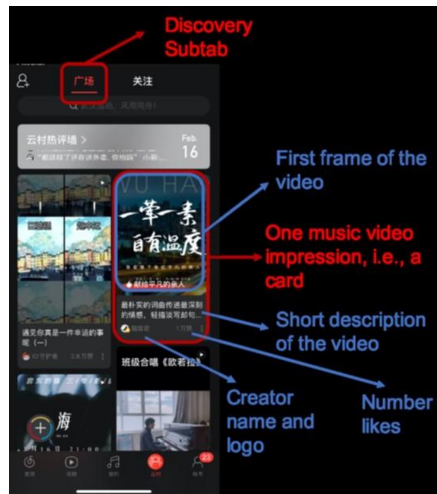


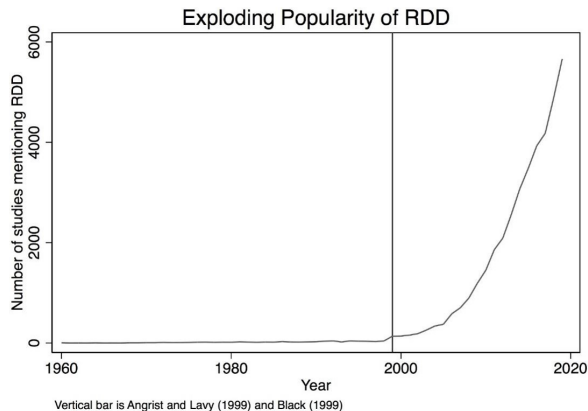
Figure 1: Anderson and Magruder (2012)

RDD in recommender systems

- **Running variable:** Algorithmic score
- **Cutoff:** Exposure threshold
- **Treatment:** In recommendation page
- **Outcome:** Engagement



The origin of RD and its growth in recent years



The first study to use RD was Thistlethwaite and Campbell (1960) on merits scores and academic outcomes.

RD is now one of the central designs for causal inference.

Figure 2: Cunningham (2021)

When a cutoff is not enough: the Russell index case

Firms near the Russell 1000/Russell 2000 cutoff appear to offer an attractive RD setting.

Lesson

A threshold does not automatically create a valid RD design.

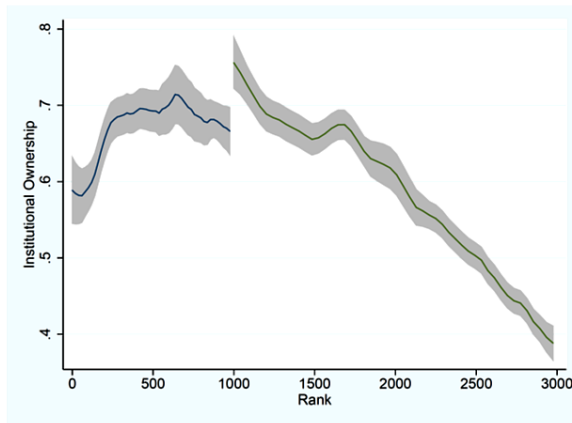


Figure 3: Bird and Karolyi (2017, retracted)

From examples to formal RD

What the examples teach us

RDD is powerful because many real-world decisions are made by rules, thresholds, scores, and rankings.

What the Russell example warns us

A visible cutoff is not enough. We need to know when the comparison just below and just above the cutoff has a causal interpretation.

Next: What exactly is the RD estimand, and what assumptions identify it?

① Getting a little “jumpy”: the basics of RDD

② RDD: A Formal Examination

The data-generating process of RDD

Local experiment perspective

Potential-outcomes perspective

③ Estimation Issues for RDD

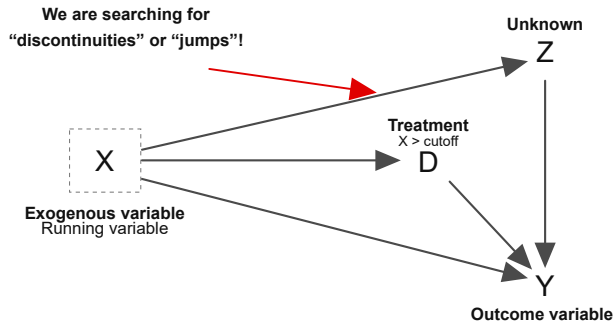
④ RDD Validation

⑤ Other variants of RDD

⑥ Slides for R Notebook

- ## 6 Slides for R Notebook

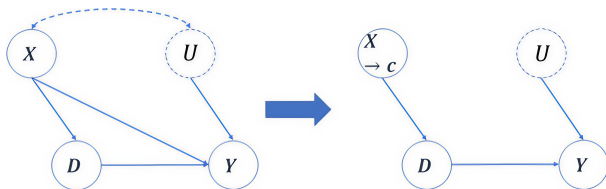
The logic of RDD



To study the effects of treatment on outcomes - **confounders exist!**

Knowledge about treatment assignment - **running variable and cutoff!**

The DAG representation of RDD

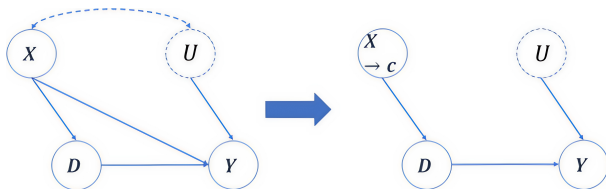


The treatment D is assigned based on the running variable X ($X \rightarrow D$).

X is likely to be associated with the confounders U ($X \leftrightarrow Z$).

For the DAG on the left, $P(Y | D)$ cannot be identified due to unblocked paths.

The DAG representation of RDD



The magic happens at the limit or the cutoff c .

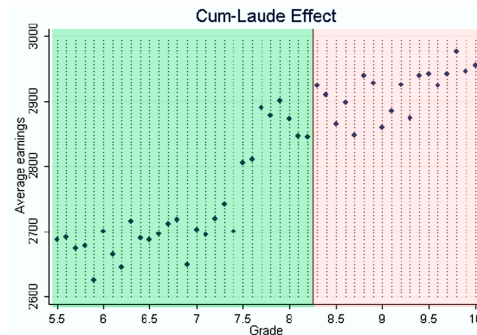
At the cutoff c , the treatment D only depends on c .

Therefore, X is excluded or the path $X \rightarrow Y$ is blocked by the cutoff c .

For the DAG on the right, $P(Y | D)$ is identified as the back-door paths between D and Y are blocked.

Data-generating process for RDD

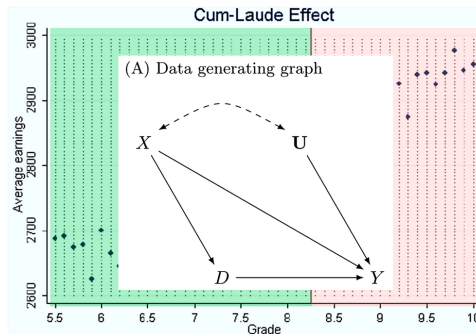
Focus on a neighborhood close to the cutoff point.



(a) Full Data

Data-generating process for RDD

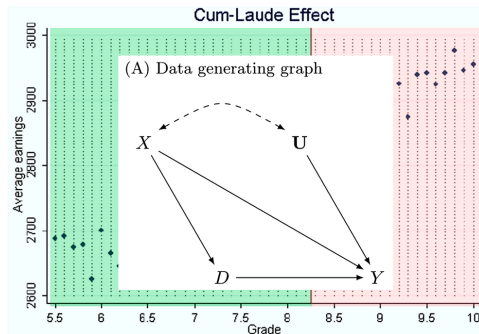
Focus on a neighborhood close to the cutoff point.



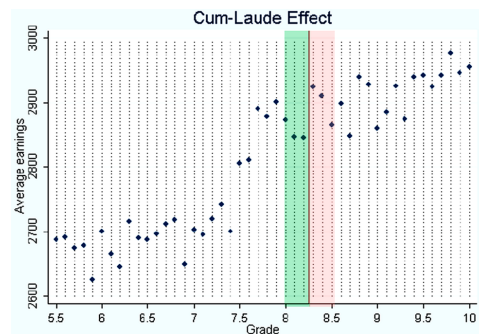
(b) Full Data

Data-generating process for RDD

Focus on a neighborhood close to the cutoff point.



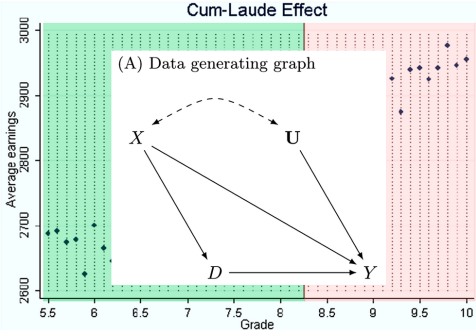
(c) Full Data



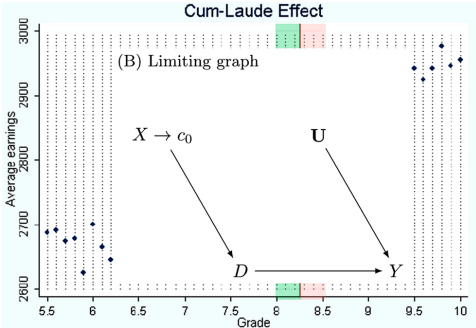
(d) RDD Data

Data-generating process for RDD

Focus on a neighborhood close to the cutoff point.



(e) Full Data



(f) RDD Data

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Local experiment perspective

Potential-outcomes perspective

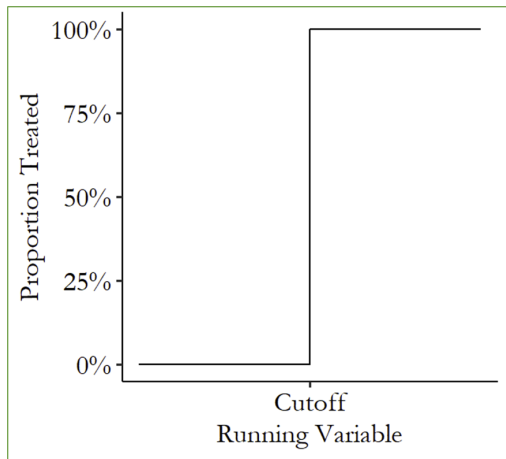
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The CRE and RDD connection



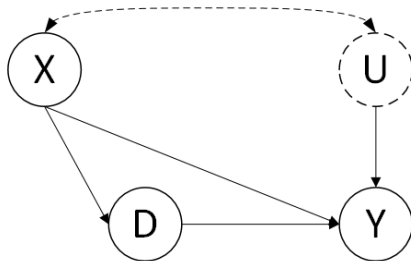
A CRE is a special case of RDD:

- Running variable: $X \sim U(0, 1)$.
- Cutoff: $c = 0.5$.
- Treatment assignment:

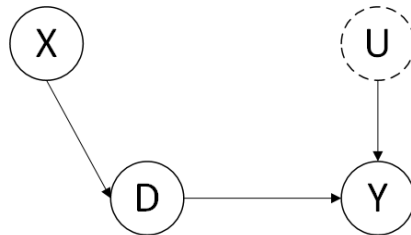
$$D = \begin{cases} 1 & \text{if } X \geq c \\ 0 & \text{if } X < c \end{cases}$$

The CRE and RDD connection

A CRE is a special case of RDD.

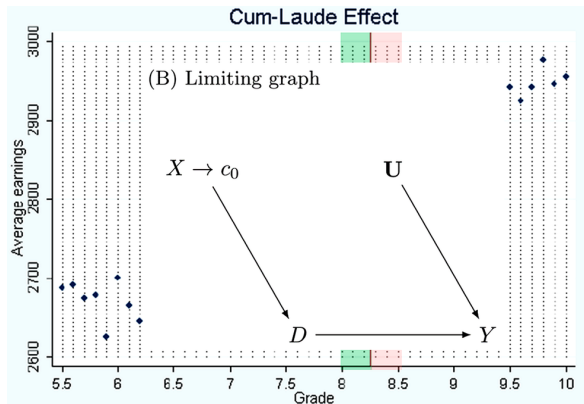


(g) DAG for RDD; $\rho(X, U) \neq 0$

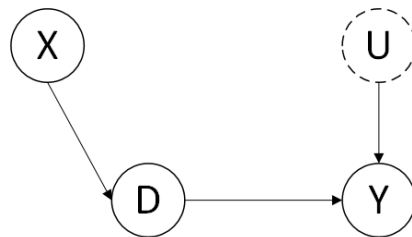


(h) DAG for CRE; $\rho(X, U) = 0$

RDD as a local randomized experiment



(i) Limiting DAG for RDD

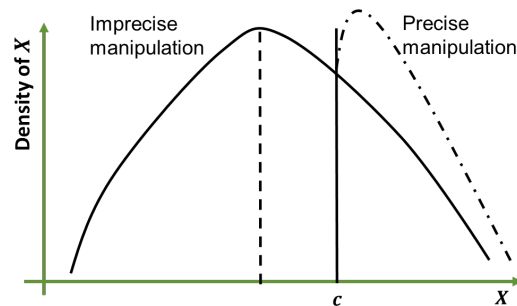


(j) DAG for CRE

RDD as a local randomized experiment

What assumption is needed for RDD as a local experiment?

- The local treatment D should be “as good as random”.
- **No manipulation:** people do not have precise control over the running variable.



RDD Estimation with no manipulation assumption

Algorithm 1: Local Experiment RDD Estimation

Input: Running variable X_i , Treatment variable D_i , Cutoff c , Outcome Y_i

Output: Estimated treatment effect $\tau_{RDD} = \mathbb{E}(Y_i^1 \mid X_i = c) - \mathbb{E}(Y_i^0 \mid X_i = c)$

- 1 Select window $(c - h_0, c + h_1)$, where $h_0, h_1 > 0$;
- 2 Trim data to include only observations with $X_i \in (c - h_0, c + h_1)$;
- 3 Estimate treatment effect with difference-in-means:

$$\hat{\tau}_{RDD} = \frac{1}{n_T} \sum_{i \in T} Y_i - \frac{1}{n_C} \sum_{i \in C} Y_i$$

where $T = \{i \mid X_i \in (c, c + h_1)\}$ and $C = \{i \mid X_i \in (c - h_0, c)\}$.

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Local experiment perspective

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Identification with potential outcomes perspective

Suppose we have a running variable X_i with the cutoff c , the treatment D_i and the outcome Y_i .

- Given the setup of RDD, the treatment assignment is:

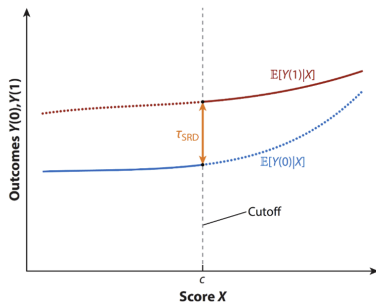
$$P(D_i = 1 \mid X_i \geq c) = 1$$

$$P(D_i = 1 \mid X_i < c) = 0$$

- The potential outcome and the observed outcome are:

$$Y_i = Y_i^1 D_i + Y_i^0 (1 - D_i)$$

Identification with potential outcomes perspective



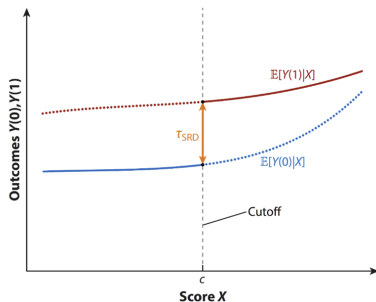
The treatment effect is identified at the cutoff

$$\tau_{RDD} = E(Y_i^1 - Y_i^0 \mid X_i = c)$$

To define τ_{RDD} in terms of limits:

$$\tau_{RDD} = \lim_{X_i \downarrow c} \mathbb{E}(Y_i^1 \mid X_i = x) - \lim_{X_i \uparrow c} \mathbb{E}(Y_i^0 \mid X_i = x)$$

Identification with potential outcomes perspective



$$\tau_{RDD} = \lim_{X_i \downarrow c} E(Y_i^1 | X_i = x) - \lim_{X_i \uparrow c} E(Y_i^0 | X_i = x)$$

For τ_{RDD} to exist, the two limits must exist.

A sufficient condition: **both $E(Y_i^1 | X_i = x)$ and $E(Y_i^0 | X_i = x)$ are continuous.**

Understanding the continuity assumption

- **Conditional unconfoundedness:** D_i is unconfounded conditional on X_i .

$$Y_i^0, Y_i^1 \perp D_i | X_i$$

- **Overlap assumption:** $\forall X_i$, the propensity score $P(D_i | X_i) \in (0, 1)$.
- But for RDD, we have:

$$P(D_i = 1 | X_i \geq c) = 1$$

$$P(D_i = 1 | X_i < c) = 0$$

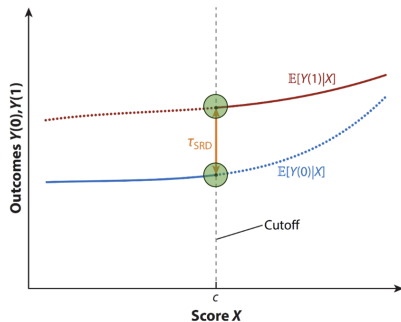
Understanding the continuity assumption

So, the overlap assumption is violated. **What are the consequences?**

$$\tau_{RDD} = \underbrace{\lim_{X_i \downarrow c} E(Y_i^1 | X_i = x)}_{P(D_i=1|X_i > c)=1} - \underbrace{\lim_{X_i \uparrow c} E(Y_i^0 | X_i = x)}_{P(D_i=1|X_i \leq c)=0}$$

- Only $\lim_{X_i \downarrow c} E(Y_i^1 | X_i = x)$ is observed, as $P(D_i = 1 | X_i > c) = 1$.
- Extra assumptions are needed!

Understanding the continuity assumption



We need to somehow “infer”
 $\lim_{X_i \uparrow c} E(Y_i^0 | X_i = x).$

Learn this function

$$E(Y_i^0 | X_i < c) = \sum_{X_i < c} Y_i^0 P(Y_i^0 | X_i).$$

Extrapolate to $E(Y_i^0 | X_i = c)$ under the **continuity assumption**.

Estimation procedure with continuity assumption

Algorithm 2: RD estimation with continuity assumption

Input: Running variable X_i , Treatment variable D_i , Cutoff c , Outcome Y_i

Output: Estimated treatment effect τ_{RDD}

- 1 Learn $\mathbb{E}(Y_i^0 \mid X_i < c)$ using observations with $X_i < c$;
- 2 Learn $\mathbb{E}(Y_i^1 \mid X_i \geq c)$ using observations with $X_i \geq c$;
- 3 Compute $\mathbb{E}(Y_i^0 \mid X_i = c)$ using the learned function from Step 1;
- 4 Compute $\mathbb{E}(Y_i^1 \mid X_i = c)$ using the learned function from Step 2;
- 5 Compute treatment effect:

$$\tau_{RDD} = \mathbb{E}(Y_i^1 \mid X_i = c) - \mathbb{E}(Y_i^0 \mid X_i = c)$$

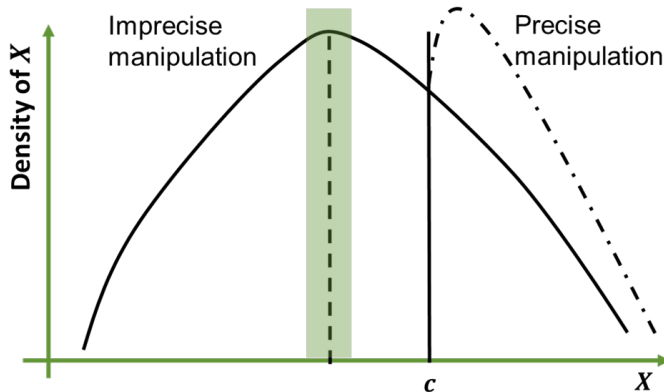
A synthesis of two perspectives

Continuity



No Manipulation^a

^aSee the formal proof in Section 4.2.4 of the monograph.

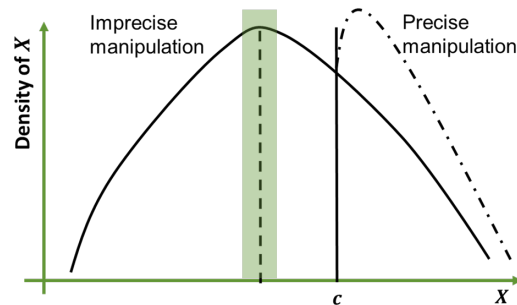


- ① Getting a little “jumpy”: the basics of RDD
- ② RDD: A Formal Examination
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Which estimation procedure?

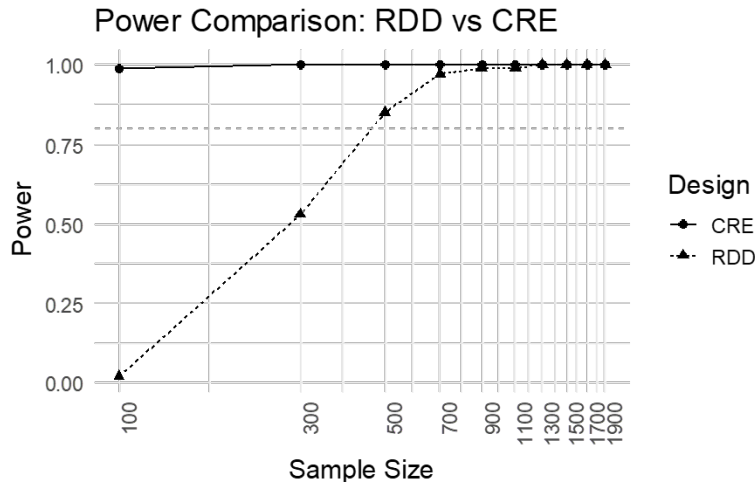
In practice, the procedure with potential outcomes is commonly used.

Why? **Low power of RDD** due to data trimming - a close neighborhood around the cutoff.



The power of RDD is lower than CRE

Low power of RDD due to data trimming.



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Predict conditional expectations

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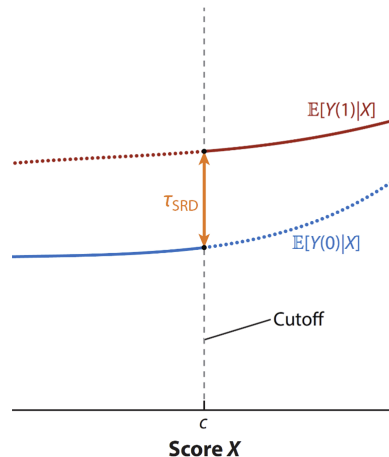
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How to predict conditional expectations?

Under the continuity assumption, we can estimate the conditional expectation function.

- Using data above the cutoff ($X \geq c$) to estimate $E(Y^1|X)$.
- Using data below the cutoff ($X < c$) to estimate $E(Y^0|X)$.



Parametric models for predicting expectations

Parametric models for $E(Y^1|X)$ and $E(Y^0|X)$.

- Linear:

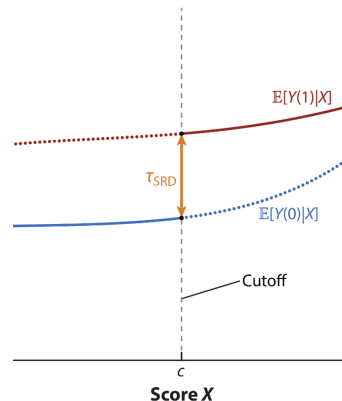
$$Y_i = \alpha + \beta D_i + \gamma X_i + e_i.$$

- With interaction:

$$Y_i = \alpha + \beta D_i + \gamma X_i + \theta D_i X_i + e_i.$$

- Polynomial:

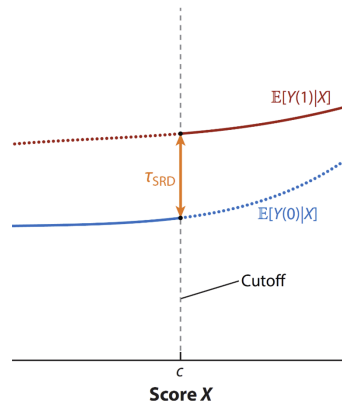
$$Y_i = \alpha + \beta D_i + \gamma_1 X_i + \gamma_2 X_i^2 + \theta_1 D_i X_i + \theta_2 D_i X_i^2 + e_i.$$



A side note about the order of the polynomials

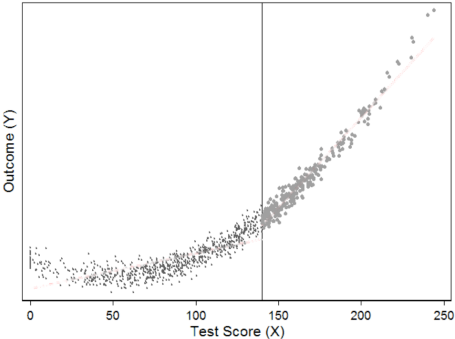
Gelman and Imbens (2019) pointed out several problems and recommend using only polynomials up to the second degree (quadratic).

- Polynomial estimation can be noisy with polynomials of higher order.
- Estimates can be sensitive to the degree of the polynomial.
- Possible bias from higher-order polynomials - Confidence intervals with bad coverage.

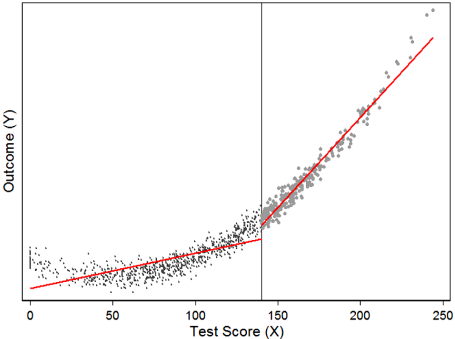


Specification errors with parametric models

In the graph, there is no jump, but a linear specification leads to “pseudo-jumps.”



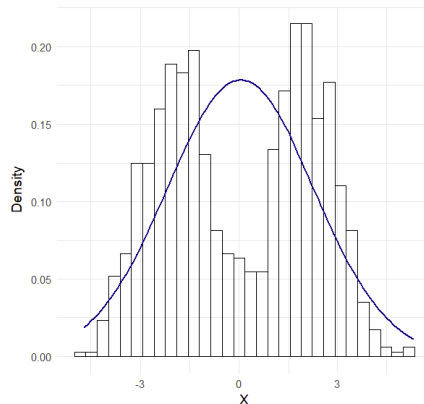
(k) Non-linear $E(Y|X)$



(l) Linear model spec

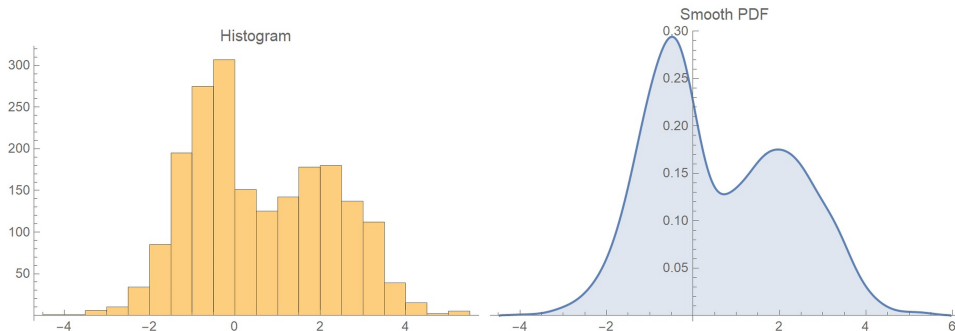
Non-parametric approach to predict expectations

- Every parametric model makes assumptions of the functional form - **prone to spec errors**.
- Instead, why not make **no functional assumptions** and let the data inform us?
- For this, adopt the **non-parametric approach**.



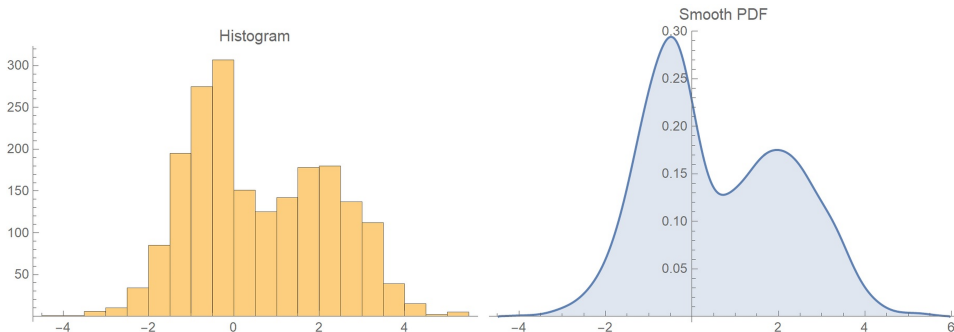
A glimpse at non-parametric estimation

- In simple cases, it's straightforward, e.g., estimate the mean $\hat{E}(X) = N^{-1} \sum X_i$ (no parameters).
- For discrete variables, the probability density function is just $p_1(X) = N_1^{-1} \sum D(X_i \in C_1)$.



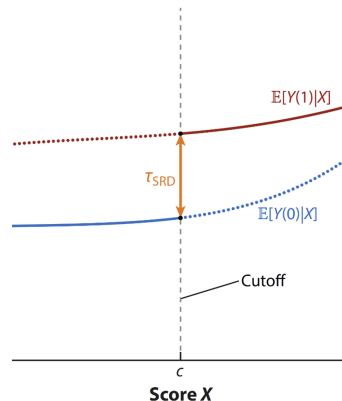
A glimpse at non-parametric estimation

- For continuous variable, we use **binning** as in producing a histogram.
- To further use small **kernels** (a smoothing function) to smooth out the histogram.



Summary: predicting expectation functions

- The true functional form is unknown - **no ground truth**.
- **No guarantee** that a non-parametric model is better, or a complex model is better.
- In practice, we often try **different models** to show the robustness.
- However, if the “**jump**” is **big**, the functional forms probably do not matter.



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Predict conditional expectations

Select appropriate samples

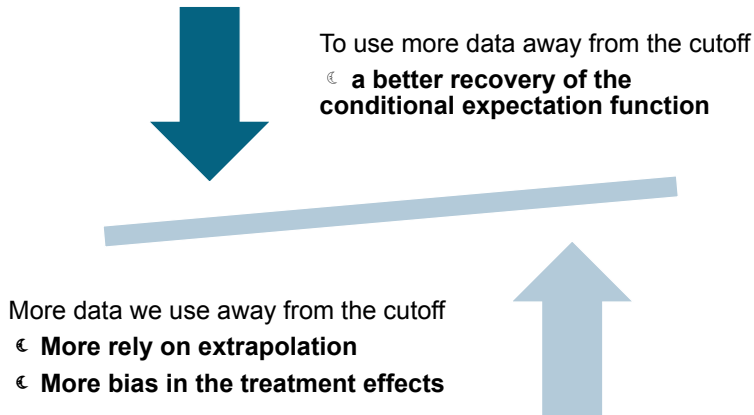
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What data to use?

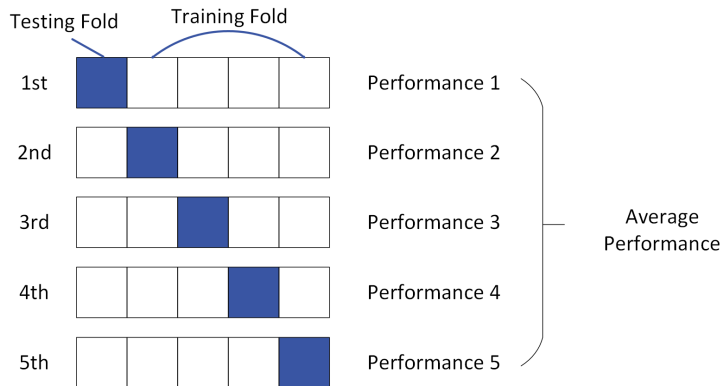
Key consideration: **the variance-bias tradeoff.**



How to select data?

Use cross-validation to choose “bandwidth.”

Balancing in-sample vs. out-of-sample fitting.



The cross-validation for RDD estimation

Algorithm 3: RDD Bandwidth Selection via Cross-Validation

Input: Grid of bandwidth parameters $\{\rho_j\}_{j=1}^M$, Number of data folds K

Output: Optimal bandwidth ρ^*

- 1 Randomly split the full data into K folds;
- 2 **foreach** ρ_j in $\{\rho_1, \rho_2, \dots, \rho_M\}$ **do**
- 3 Set bandwidth $h_j = \rho_j \cdot SD(X_i)$;
- 4 **for** $k = 1$ **to** K **do**
- 5 Keep observations satisfying $|X_i - c| \leq h_j$ in both the training and validation sets;
- 6 Fit $E[Y_i | X_i, D_i]$ using training sets $K \setminus \{k\}$;
- 7 Compute performance (e.g., MSE) on validation set k ;
- 8 Compute average performance over K folds for h_j ;
- 9 Select ρ^* with the best performance;

Summary: the general estimation procedure for RDD



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Examining the continuity assumption

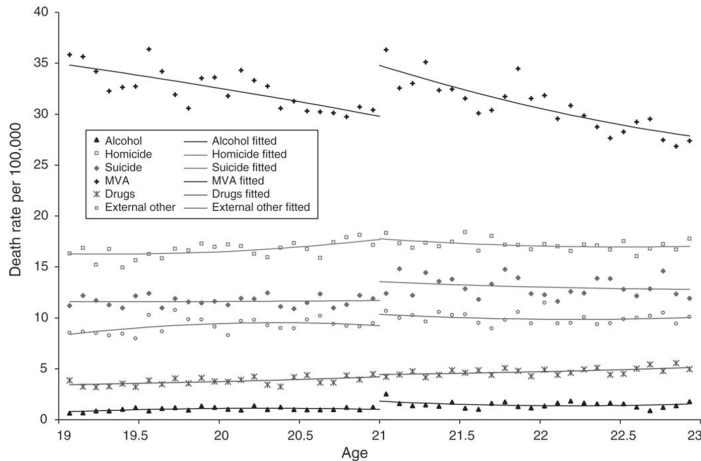


Figure 4: Carpenter and Dobkin (2009)

We do not observe the counterfactual outcomes.

Use outcomes that should be / not be affected by the treatment as “consistency tests.”^a

^aMore on placebo / consistency tests in the last session.

Examining the continuity assumption

Why don't we examine the distribution of the running variable X_i directly?

Expect no discontinuity around the cutoff ("no manipulation") - McCrary density test.

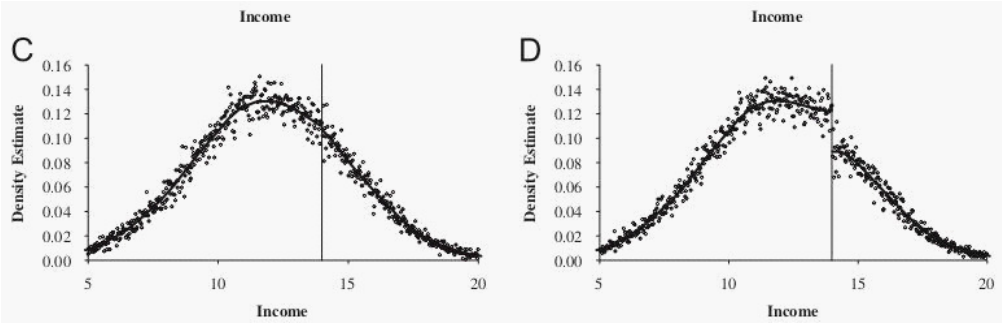
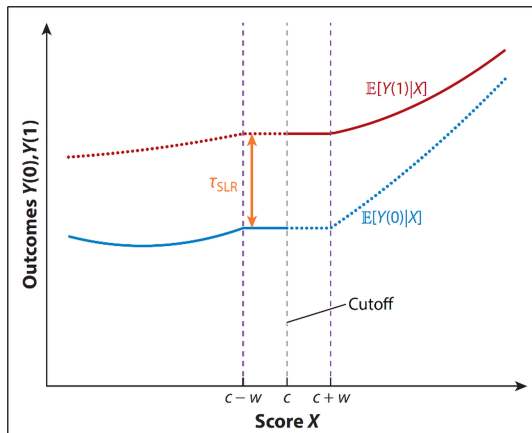


Figure 5: McCrary (2008)

Examine the continuity assumption



The idea of “local experiments.”

Implication: in the close-neighborhood of the cutoff c , the assignment of treatment should be “as-if” random.

Balance tests for RDD

If we have local randomization, the characteristics of people should be **balanced above and below the cutoff**.

Balance test: covariates are balanced around the cutoff.

Variable	Std. Mean Difference	P-value
<i>Age</i>		
25 - 44	0.018	0.849
45 - 64	-0.018	0.849
65 ≥	-0.087	0.360
<i>Education Level</i>		
Primary	0.078	0.409
Secondary	0.058	0.542
Post-secondary	-0.151	0.111
<i>Household Wealth Index</i>		
Q1	-0.056	0.538
Q2	0.00	1.000
Q4	0.142	0.135
Q5	-0.154	0.104
<i>Region</i>		
Northern	-0.155	0.103
Middle	0.158	0.102
<i>Other</i>		
Household size	-0.007	0.942
Household size squared	-0.051	0.599
Overweight (25 ≤ BMI ≤ 29.9)	0.00	1.000
Renewed insurance at least once	0.111	0.241
Female	0.036	0.701
Married	0.051	0.593
Urban	0.032	0.737
At least one child in the household	0.00	1.000

Two perspectives on RDD identification

Perspective	Core intuition	Key assumption	Empirical checks
Continuity	Potential outcomes evolve smoothly through the cutoff	No discontinuous confounders at c	Placebo outcomes; covariates
Local randomization	Near the cutoff, treatment is as-if random	No precise manipulation of X	Density test; balance tests

Both perspectives ask whether units just below/above the cutoff are credible counterfactuals.

Interpreting RD treatment effects

**Important: how to
interpret τ_{RDD} ?**



In RDD, treatment effects are local average treatment effects or LATE.



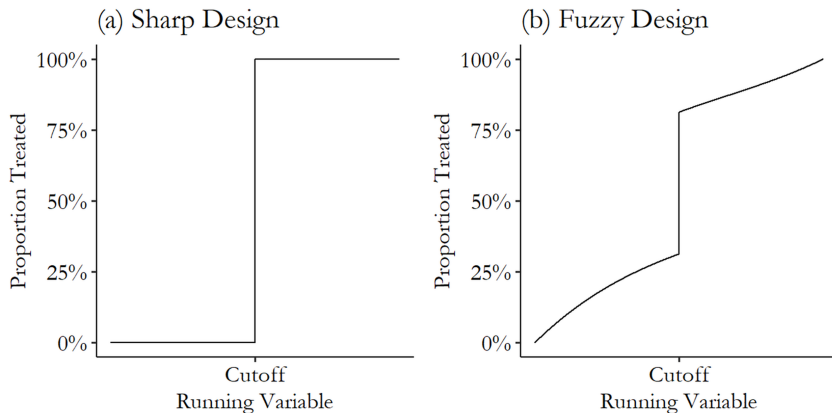
Not average treatment effect, but rather the effect of getting the treatment for units that were close to the cutoff.



This is the price we pay for a non-random assignment.

- 1 Getting a little “jumpy”: the basics of RDD
- 2 RDD: A Formal Examination
- 3 Estimation Issues for RDD
- 4 RDD Validation
- 5 Other variants of RDD**
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Fuzzy RDD

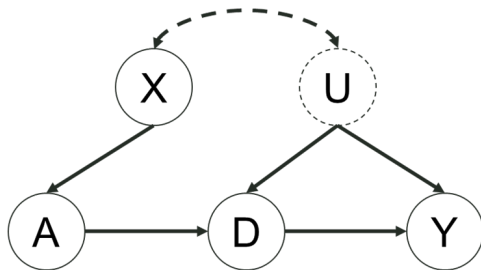


Fuzzy RDD: the treatment has **non-compliance**.

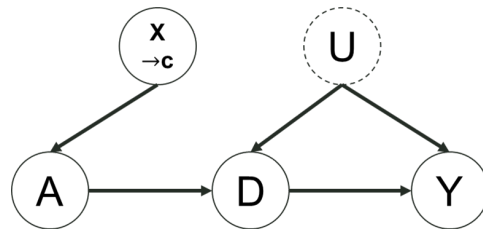
Figure 6: Comparing Sharp vs. Fuzzy RDD

DAGs for fuzzy RDD

- The assignment A depends on the cutoff of the running variable.
- The actual treatment D depends on the assignment.



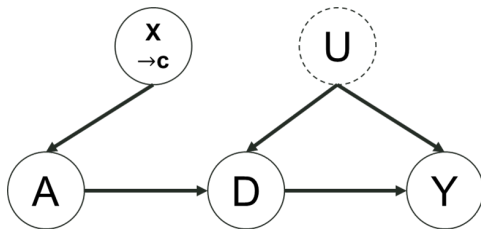
(a) Data generating DAG



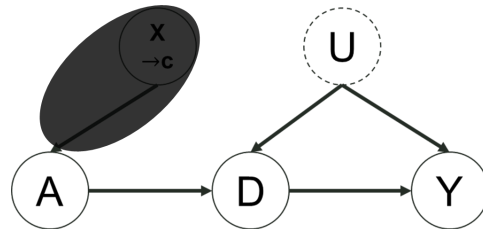
(b) Limiting DAG

DAGs for fuzzy RDD

- In the limit $X \rightarrow c$, the DAG becomes just like in the non-compliance setting.
- Assignment A becomes an instrument for the treatment D .



(c) Limiting DAG



(d) IV estimation

Regression kink design (RKD)

- RDD centered around a **discontinuous jump** in the treatment (binary).
- Many policy tools have shifts in **the treatment intensity** based on the running variable, rather than jumps.



Income taxes



Usage-based pricing

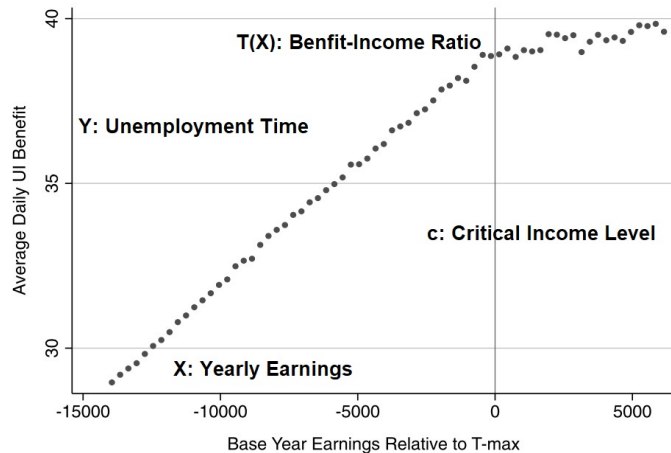


Tiered loyalty program

Figure 7: Examples of RKD

Basic elements of RKD

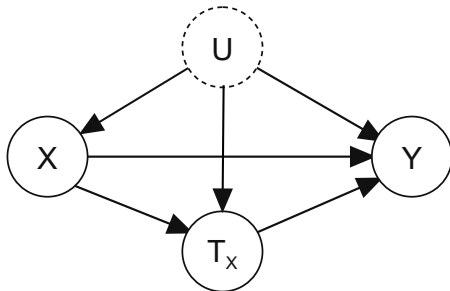
X	cause
Y	outcome
$T(X)$	treatment
c	cutoff



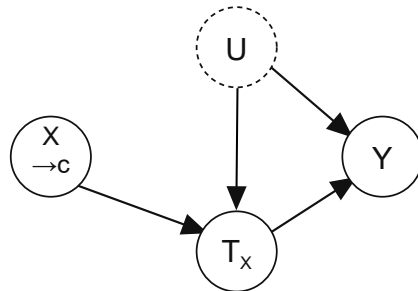
Identification for RKD

The conditional expectations of the potential outcomes $E(Y|X)$ are **continuously differentiable** at the cutoff c .

Implications: **Only the treatment rule leads to the change in slopes.**



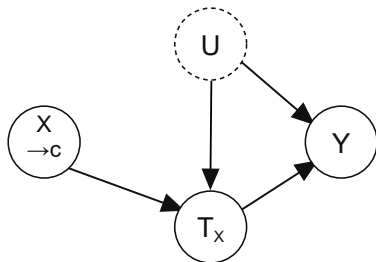
(a) Data generating DAG



(b) Limiting DAG

Identification results of RKD

The local marginal effect of treatment intensity $T(X)$ on outcome Y at the cutoff c .



$$\tau_{RKD} = \frac{\Delta \left(\frac{\partial \mathbb{E}[Y|X]}{\partial X} \right)}{\Delta \left(\frac{\partial T(X)}{\partial X} \right)} \Bigg|_{X=c}$$

Similar to instrumental variable estimation:

$$\tau_{RKD} = \frac{\lim_{x \downarrow c} \frac{\partial}{\partial x} \mathbb{E}[Y | X = x] - \lim_{x \uparrow c} \frac{\partial}{\partial x} \mathbb{E}[Y | X = x]}{\lim_{x \downarrow c} \frac{\partial}{\partial x} T(X) - \lim_{x \uparrow c} \frac{\partial}{\partial x} T(X)}$$

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Keywords auction on Google

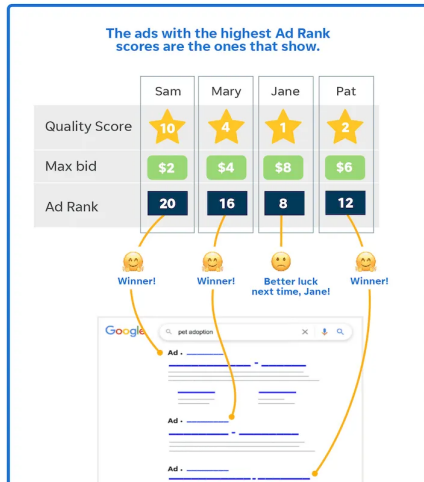


Objective: **to measure the value of positions.**

Observations:

- The position is determined by AdRank.
- For two positions j and $j + 1$, we must have $\text{AdRank}_j - \text{AdRank}_{j+1} \geq 0$.

Using RDD to evaluate position effects



- **Running variable:**
 $\Delta \text{AdRank}_{ij} = \text{AdRank}_i - \text{AdRank}_j$.
- **Cutoff:** if $\Delta \text{AdRank}_{ij} > 0$.
- **Treatment:** Rank at position $k + 1$ vs. k .
- **Outcome:** Revenue or other variables.

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